

Modulus, Argument form

Key Points

The modulus and argument of a complex number, $z = x + yi$ are

$$|z| = \sqrt{x^2 + y^2}$$

$$\text{Arg}(z) = \tan^{-1}\left(\frac{y}{x}\right)$$

With care taken when working out the argument when $x \leq 0$

The modulus argument form is

$$z = |z|(\cos(\text{Arg}(z)) + i \sin(\text{Arg}(z)))$$

Basic Skills

Q1. Work out the modulus of the following complex numbers

- a) $z = 8 + 6i$
- b) $z = 4 - i$
- c) $z = 4i$

Q2. Work out the argument of the following complex numbers

- a) $z = 4 + 4i$
- b) $z = 2i - 1$
- c) $z = -4i$
- d) $z = -5$

Competency Skills

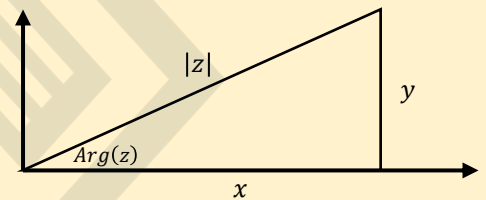
Q3. Explain why $z = x + yi = |z|(\cos(\text{Arg}(z)) + i \sin(\text{Arg}(z)))$ holds true by using the triangle

Q4. Write the following complex numbers in modulus, argument form

- a) $z = 4 + 4i$
- b) $z = 3 - 4i$
- c) $z = i - \sqrt{3}$
- d) $z = -i - \sqrt{3}$

Q5. For the complex number $z = \frac{1}{2} - 2i$

- a) Draw the complex number z indicating the modulus and argument
- b) Draw the complex number $2z$ indicating the modulus and argument
- c) Draw the complex number $z + 2i$ indicating the modulus and argument



Advanced Skills

Q6. For the complex numbers $z = 3 + 4i, w = i - 2$

- a) Draw the complex number z indicating the modulus and argument
- b) Draw the complex number w indicating the modulus and argument
- c) Find the modulus and argument for z^2
- d) Draw the complex number z^2 indicating the modulus and argument
- e) Find zw in modulus, argument form
- f) Draw the complex number zw indicating the modulus and argument
- g) Using e, f) determine if $|zw| = |z||w|$ and $\text{Arg}(zw) = \text{Arg}(z) + \text{Arg}(w)$ are true

Extension

Q7. Is it true that $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$ and $\text{Arg}\left(\frac{z}{w}\right) = \text{Arg}(z) - \text{Arg}(w)$

Hint: use Q6.g) as a guideline, also drawing a diagram will help with visualisation